Zachary Bezemek

(248)-978-5905|zwb@duke.edu|scholars.duke.edu/person/zachary.bezemek

ACADEMIC POSITIONS

Duke University | Durham, NC *August 2023 – Present*

- ***Phillip Griffith Assistant Research Professor of Mathematics***

EDUCATION

Boston University | Boston, MA *September 2018 – May 2023*

- **PhD, Mathematics**
- Advisor: Prof. Konstantinos Spiliopoulos
- Thesis: *Interacting particle systems in multiscale environments: asymptotic analysis*

Michigan State University | East Lansing, MI *September 2014 – May 2018*

- **Bachelor of Science, Advanced Mathematics**
- Additional Major in Japanese, Honors College

CERTIFICATIONS

The Erdős Institute | Virtual *May 2024 – June 2024*

- **Data Science Boot Camp**
- Project: Geo-Locator – A classifying model that can predict in which out of 17 cities a picture was taken
- Top 5 project out of 58 teams
- GitHub: https://github.com/hochfilzer/geo-locator

The Erdős Institute | Virtual *June 2024 – September 2024*

- **Deep Learning Boot Camp**
- Project: RivusVox Editor – the world's first near-live zero-shot adaptive speech editing system
- First-place project out of 26 teams
- GitHub: https://github.com/bezemekz/RivusVoxEditor

CONFERENCE PAPERS/PREPRINTS

Planner Aware Path Learning in Diffusion Language Models Training (arXiv:2509.23405) *September 2025*

Joint work with Fred Peng, Jarrid Rector-Brooks, Shuibai Zhang, Anru R. Zhang, Michael Bronstein, Avishek Joey Bose, and Alexander Tong.

- First author. ICLR 2026 Oral

Path Planning for Masked Diffusion Model Sampling (arXiv:2502.03540) *Februrary 2025*

Joint work with Fred Peng, Sawan Patel, Jarrid Rector-Brooks, Sherwood Yao, Avishek Bose, Alexander Tong, and Pranam Chatterjee.

- First author. Long Paper Outstanding Paper Award ICLR Deep Generative Model in Machine Learning Workshop 2025

PUBLICATIONS

Importance Sampling for the Empirical Measure of Weakly Interacting Diffusions (arXiv:2210.08591) *2024*

Joint work with Max Heldman. Applied Mathematics & Optimization Vol 89, No.7

Moderate deviations for fully coupled multiscale weakly interacting particle systems (arXiv:2202.08403) *2023*

Joint work with Konstantinos Spiliopoulos. Stochastics and Partial Differential Equations: Analysis and Computations. Vol. 12, pp.1265–1373

Large deviations for interacting multiscale particle systems (arXiv:2011.03032) *2023*

Joint work with Konstantinos Spiliopoulos. Stochastic Processes and their Applications Vol. 115., No.1, pp.27-108

Rate of homogenization for fully-coupled McKean-Vlasov SDEs (arXiv:2202.07753) *2023*

Joint work with Konstantinos Spiliopoulos. Stochastics and Dynamics Vol. 23, No.2, pp.1-65

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CONFERENCES, PANELS, EVENTS

Oral Presentation, ICLR *April 2026*
Poster Presentation, Algorithmic Advances in Generative AI Conference at Duke University *February 2026*
Invited Speaker, INFORMS APS 2025 *July 2025*
Invited Speaker, Cradle Tech Talk (Zurich) *March 2025*
Participant, Institute for the Quantitative Study of Inclusion, Diversity, and Equity Datathon for Justice *October 2024*
Poster Presentation, Seminar on Stochastic Processes, Rice University *March 2024*
Invited Speaker, Probability Seminar at Duke *February 2023/September 2023*
Invited Speaker, ICIAM 2023 Minisymposium on Scaling Limits of Interacting Particle Systems *August 2023*
Invited Speaker, Probability and Statistics Seminar at KTH *May 2023*
Invited Speaker, Brin MRC Workshop on Rare Events *February 2023*
Invited Speaker, Probability Seminar at Duke *February 2023*
Invited Speaker, Probability/Math Finance Seminar at CMU *January 2023*
Invited Speaker, Probability and Statistics Seminar and Math Finance Colloquium at USC *January 2023*
Invited Speaker, FOR 2402 SPDE Seminar at TU Berlin *January 2023*
Speaker, Graduate Student Probability Conference at Madison *September 2022*
Invited Speaker, Brown / BU / UMass Dynamics & PDE Seminar *April 2022*
Organizer, CISE Graduate Student Workshop 8.0/9.0 *April 2022/January 2023*
Lightning Talk, CRM Workshop on Mean Field Games *April 2022*
Graduate Student Panelist, NSF REU Site in Mathematical Analysis and Applications UMich Dearborn *June 2021*
Speaker, Triangle Area Graduate Mathematics Conference *December 2020*
Invited Speaker, Boston University Probability and Statistics Seminar *September 2020*

TEACHING AND MENTORING EXPERIENCE AND AWARDS

The Erdős Institute | Virtual
Data Science Bootcamp Mentor for a Top 5 Project (out of 34 teams) *Spring 2025*
Mathematics Department, Duke University | Durham, North Carolina
Instructor – MATH 541/STAT 621 (Applied Stochastic Processes) *Fall 2024, Spring 2026*
Instructor – MATH 219/ MATH 719 (Multivariable Calculus) *Fall 2025*
Awardee – Lewis Blake Award for Excellence in Teaching *Fall 2024 – Spring 2025*
Advisor – Research Training Group: Analysis, Probability, PDE, Computation Research Experience for Undergraduates *Spring 2025*
Instructor – MATH 340/ STAT 231/ MATH 740 (Advanced Intro to Probability) *Spring 2025*
Instructor – MATH 431/MATH 731 (Real Analysis) *Spring 2024*
Instructor – MATH 230/ STAT 230/ MATH 730 (Intro to Probability) *Fall 2023*

Mathematics Department, Boston University | Boston, Massachusetts
Teaching Fellow – MA583 (Intro to Stochastic Processes) *Spring 2022*
Directed Reading Program Mentor – Real Analysis *Spring 2022*
Teaching Fellow – MA 123/124 (Calc I/II) *Fall 2019, Fall 2020, Spring 2021, Fall 2021*
Mentor – Departmental Undergraduate Peer Mentorship Program *Fall 2020, Spring 2021*
Teaching Fellow – MA116 (Stats II) *Spring 2020*
Instructor – MA 242 (Linear Algebra) *Summer 2019*

REVIEWS

Annales de l'Institut Henri Poincaré
Potential Analysis
Applied Probability Journals
The Annals of Probability

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The Annals of Applied Probability
SIAM Journal on Control and Optimization

UNDERGRADUATE RESEARCH EXPERIENCE

NSF REU Site in Mathematical Analysis and Applications – Prof. Hyejin Kim *May 2017 – August 2017*
Mathematics and Statistics Department, University of Michigan-Dearborn | Dearborn, Michigan
Mathematics Exchange Program Research Assistant – Prof. Keith Promislow *January 2016 – May 2017*
Mathematics Department, Michigan State University | East Lansing, Michigan
Professorial Assistant – Prof. John Deller *August 2014 – May 2016*
Speech Processing Laboratory, Michigan State University | East Lansing, Michigan

UNDERGRADUATE CONFERENCES

Poster Presentation, Joint Mathematics Meetings 2018 – MAA Student Travel Grant Awarded *January 2018*
Speaker, Mathematical Biology Institute Undergraduate Capstone Conference *August 2017*
Speaker, Summer Undergraduate Michigan Mathematics Research Conference *July 2017*
Conference Winner, MSU Undergraduate Student Mathematics Conference *Spring 2016*

UNDERGRADUATE TEACHING EXPERIENCE

Mathematics Department, Michigan State University | East Lansing, Michigan
Undergraduate Learning Assistant – Quantitative Literacy *Spring 2018*
Grader - Introduction to Analysis *Fall 2017*
Undergraduate Learning Assistant – Pre-calculus and Trigonometry *Fall 2016*
Advanced Calculus Tutor at Math and Learning Center *August 2014 – May 2018*

OTHER WORK EXPERIENCE

Programming Intern *May 2016 – August 2016*
HIROTEC America | Auburn Hills, Michigan
Ford Blue Oval Intern *August 2014 – September 2015*
Residential College of Arts and Humanities / Engineering College, Michigan State University | East Lansing, Michigan

TECHNICAL SKILLS

- Python
- JavaScript
- MATLAB
- Mathematica
- LaTeX
- Linux
- Windows
- Microsoft Office Suite
- Godot
- Ableton

LANGUAGES

- English (native)
- Japanese (conversational)